

Virtual Immersion in Underground Quarries: An Innovative Exploration of Historical and Cultural Heritage

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Figure 1: Views from the VR application : (from left to right) animated quarry worker, the Blue Lake, and interactive quarry inspector

ABSTRACT

Underground quarries, as essential elements of cultural heritage, present unique challenges in terms of mediation, both from a safety and accessibility perspective. The use of virtual reality (VR) in this field allows for immersion in complex underground environments, thus providing an innovative solution to showcase them to the general public. This document presents a virtual reality installation dedicated to underground quarries, explaining its purpose, objectives, and possibilities.

Index Terms: VR, underground, quarry, accessibility, mediation

1 INTRODUCTION

Underground quarries represent a valuable cultural heritage, showcasing human ingenuity in the extraction of natural resources throughout history. These sites, often forgotten or difficult to access, contain important remnants related to industry, geology, and history, but they also present unique challenges in terms of conservation and mediation. Indeed, the safety of visitors and the accessibility of these complex spaces require innovative solutions to enable true immersion while ensuring the preservation of the sites.

The integration of virtual reality (VR) technologies in cultural heritage has proven useful in many other contexts [1]. By offering the ability to recreate detailed and interactive underground environments, VR might allow for overcoming the physical limitations of exploring these quarries. This technology might thus democratise access to these spaces while offering an interactive and immersive experience with added information, capable of attracting a wide audience.

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This article aims to present such a VR installation dedicated to underground quarries in the city of Lezennes (France), designed by PIRVI (Pôle de compétences Interactions Réalité Virtuelle et Images / Skill Center for Interactions, Virtual Reality and Images, CRISTAL, Université de Lille) and LGCgE (Laboratoire de Génie Civil et géo-Environnement / Civil Engineering and Geo-Environment Laboratory).

The goal of this project was to provide a virtual version of the quarries for conservation and mediation purposes. On the mediation side, this virtual reconstruction is to be presented in regional exhibitions for schools and the general public, but it may also inform the design of similar experiences for cities around Lezennes, in the north of France and Belgium.

In this paper, we first present the Lezennes quarries. We then analyse the objectives of this installation, focusing on its content, the possibilities offered by VR, and its implications in terms of accessibility.

2 DIGITISING LEZENNES UNDERGROUND QUARRIES

The European Metropolis of Lille (comprising 95 municipalities, including 12 with quarries) has nearly 160 registered quarries, which represent 4.5 million cubic meters of void over an area of 181 hectares. These quarries constitute a significant cultural heritage, with geological and historical value as well as architectural and social, yet they remain largely under-exploited. For instance, numerous inscriptions can be found, ranging from their beginnings (12th century) to the present day.

In order to highlight this heritage, PIRVI, an engineering platform specialised in VR, has partnered with geologists of LGCgE, as well as the city council of Lezennes, on several projects related to underground quarries: degradation monitoring, training, and mediation. One of these projects, supported by the Call for Expressions of Interest (AMI) *Sciences Infusent* from the University of Lille, led to the creation of an exhibition on underground quarries, featuring a VR application. The goal of this exhibition was to immerse participants, who cannot visit the quarries for safety or accessibility reasons, in a virtual twin and introduce them to the history, dangers,

jobs and uses of the quarries.

2.1 Lezennes Quarries

The village of Lezennes (about 3,500 inhabitants) has the most extended area of underground quarries (about 70 ha) of the European Metropolis of Lille. Extraction probably began during the 11th century and ended in the middle of the 19th century [2, 3]. The rock mined was a white chalk of Coniacian to Santonian age (end of the Cretaceous period, about 90 My). The material was mainly used for the construction of many buildings of the Lille metropolitan area (houses, farms as well as prestigious buildings of the City of Lille such as the medieval city fortifications, the XVIIth century citadel, several palaces, churches, ...). Additionally the carbonaceous rock was exploited during the 19th and 20th centuries to produce lime in the so-called “*catiches*” consisting of very particular bottle-shaped mining rooms. The peculiarity of the quarries is that they were used for several other purposes later on. Indeed, they were exploited from the middle of 19th century to the middle of the 20th century for vegetable productions such as mushrooms and wild chicory (“*barbe de capucin*”). Moreover the quarries served as shelters during some conflicts, particularly during the two world wars both for soldiers and civilians of the village [2, 3].

Since 1999 the Municipality of Lezennes planned to promote this heritage in collaboration with the University of Lille and the local history association C.R.H.L. (“*Cercle de Recherche Historique Lezennois*”). Indeed, since 1999, the quarries are open to public during the European Heritage Days week-end representing an influx of 400 to 500 people during two days. Moreover, the municipality has launched several actions around the theme of subterranean quarries, for instance the creation of a processional giant that symbolises the village (Isidore), a statue dedicated to quarry works, a monument at the town entrance (white wall), a stone festival ...

Whereas very attractive to the public, the underground heritage is difficult to access. The annual visit is limited to a single gallery representing about 40 meters length. Inferred risks include going down stairs, walking in a narrow gallery, gallery height not high enough, air quality (low oxygen rate), roof and wall instabilities. Additionally, access for young children and persons with reduced mobility is impossible. Thus, access to very interesting sectors, such as the famous “Blue lake”, is impossible (and forbidden by decision of the mayor) except for specialists and quarry inspectors.

The restriction of access has led to the investigation of other solutions to satisfy the public curiosity and VR represents the approach we chose to enhance historical heritage.

2.2 Digitising the quarries

3D scanning of underground quarries offers numerous benefits, both in terms of preservation and mediation. By creating an accurate and detailed digital replica of these sites, scanning allows for a faithful record of their current condition, which is essential for their long-term protection. Indeed, underground quarries are often exposed to risks of degradation, whether due to rocky material fatigue, water saturation variations in chalk, or surface human activities [5]. Through digitisation, it becomes possible to monitor the evolution of these sites and identify areas that require conservation interventions, while offering a solution to preserve them virtually without risking physical deterioration [4].

Transforming these scanned quarries into virtual environments opens up new possibilities for cultural mediation and heritage discovery. By enabling full immersion in these often difficult-to-access environments, VR democratises access to these unique sites, whether they are located underground or in prohibited areas. This approach allows for reaching a broad audience, from researchers and students to history and geology enthusiasts, while providing an interactive experience that enhances the understanding of geological, historical, and technical processes. Moreover, VR enables the

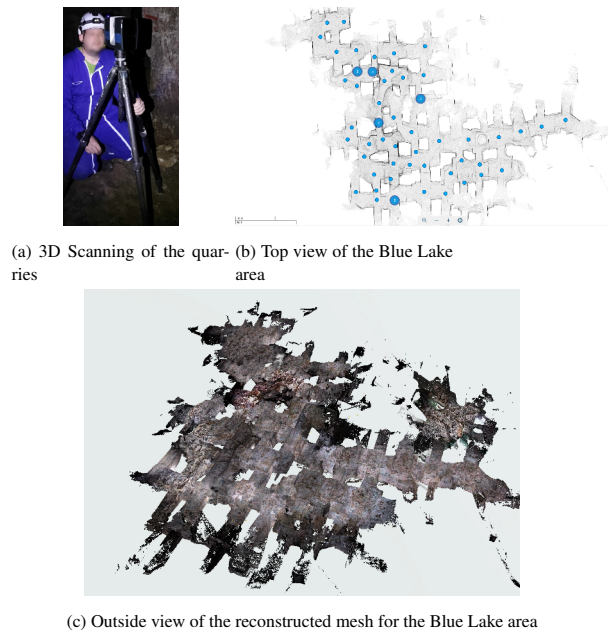


Figure 2: Digitising process of the underground quarries

exploration of these quarries in a safe manner, without the physical constraints associated with direct exploration, through various 3D navigation techniques.

That is why we conducted numerous visits to carry out more than a hundred 3D scans (using a FARO Focus, see Fig. 2a). This allowed us to create VR versions of a dozen locations, despite difficult access and fragile equipment. Each digitised site was chosen based on its geometry (large enough to navigate in the virtual environment) and its content (presence of inscriptions, geological features, tool marks, traces of various activities, etc.). For instance, Figure 2b shows the map of the *Blue Lake* area and Figure 2c depicts the resulting reconstructed mesh of this area.

3 VIRTUAL REALITY PUBLIC PRESENTATION

Once the chosen locations in the quarries were digitised, we developed a VR application in order to offer an unprecedented immersion into these often overlooked and difficult-to-access spaces. To make this experience as enriching as possible, the 3D environments representing each location are carefully arranged by themes, allowing users to explore the quarries from various perspectives : historical, geological, technical, and even patrimonial. Each thematic area provides specific information, enhancing the experience with interactive elements such as animations (characters or interactive events, such as the collapse of a gallery room to illustrate a major risk) and texts. This setup enables a personalised exploration, where each user can focus on the aspects they are interested in, while facilitating an educational and enjoyable approach for various types of audiences.

The application has also been designed with the needs of people with reduced mobility in mind, ensuring an experience that is accessible to everyone.

3.1 Selection of themes and interactions

Over time, underground quarries have been used for different reasons. We decided to present the stroll between the locations as a journey through history: from their first use, the chalk extraction activity, to their present state. Thus the visitor enters the quarry and gradually reaches different locations. The themes covered are:

1. Quarries in the Middle Ages;
2. The quarries in the 19th century;
3. Resting place for soldiers during the First World War;
4. Shelter for civilians during the Second World War;
5. Mushroom cultivation;
6. The Blue Lake (overdigging down to the water table), shown in Figure 1 centre;
7. The Scout Altar room, a peculiar worship room attributed to scout group activity;
8. Geology and paleontology;
9. Metrology and site monitoring.

Each theme is explained through text and animations. For example, in the quarry themes, the visitor can see a quarryman working, as shown in Figure 1 left, and even choose the successive movements and postures of the quarry worker; regarding scenes relating to wars, the visitor can virtually meet characters (a soldier and a female refugee) whose portraits and costumes have been designed from vintage photos.

For each location, users can point at various elements using a ray casting metaphor in order to retrieve information on the thematic, as depicted in Figure 1, right.

The transition from one theme to another is performed via a menu. The themes are organised in chronological order. We therefore made the choice not to allow the possibility of moving completely freely between them: from a theme, it is only possible to go to the previous or next theme. The menu appears when the visitor moves to the center of the physical space and appears in front of them, at the same height as their head. The menu contains a list of other themes, each represented by a name and a picture of the most iconic element in it, which get highlighted when visitors point at it.

In addition, to recreated the immersion in real quarries, we gave the visitor the opportunity to play with the light, positioned above their head like a headlamp, as can be seen in Figure 1 centre. By pressing a button, this light can be switched between different modes: off, red, low, medium and high intensity. This choice was made because many headlamps are adjustable in this way. We also added a circular flashlight pattern to replicate the way a headlamp illuminates.

3.2 Accessibility

In order to ensure accessibility for as many visitors as possible, we implemented 2 interaction modes : a seated mode and a standing mode.

The **seated mode** is designed for persons with reduced mobility (PRM) to move in the VR environment. This mode allows a visitor to move while seated; the person can use joystick to turn and go around (up and down to go forward/backward; and right and left for rotating). This mode permits visitors in wheelchair, people that cannot stand for long, or people that have difficulties in VR (like balance problems) to stay a long time in VR environment. However, this mode of navigation could cause motion-sickness and techniques to avoid it should be investigated.

The **standing mode** enables a person to walk around. The area from the real space is mapped to a space of the same size in VR, where the visitor can move. This technique allows the visitor to move freely without using any controllers.

We integrated a video tutorial in order to let visitors use the VR application without presence of a facilitator or a guide. It allows anyone to quickly grasp the interaction mechanics in just a few minutes and configure its use. The videos are accompanied by very short texts and appear only one or two at a time.

3.3 Public presentation

The application was showcased in combination with an exhibition related to the underground quarries. The VR experience was more

focused on contemplation and contains little text, whereas the exhibition included numerous posters (covering the same themes), display cases with objects, and educational games.

This exhibition began during the 2024 Heritage Days and remained open to the public for over a month (from 20/09 to 27/10). During this time, two booklets were produced : an explanatory one (explaining each theme in more detail), and a playful one (more targeted for young children).

The event was a resounding success, attracting 560 visitors in total (357 visitors on opening hours, including 10 school groups, not counting visitors during 2 special "Heritage Days"), highlighting the growing interest in immersive technologies for exploring heritage. We believe that the application not only captivated participants with its interactive nature but also helped strengthen cultural mediation by offering an innovative way to understand and preserve local history. By integrating educational tools suited to different audiences, the exhibition created a bridge between the past and the present, while sparking curiosity and enthusiasm among visitors.

4 CONCLUSION

This VR application dedicated to the exploration of underground quarries represents a contribution in the promotion of historical and cultural heritage. By offering an interactive immersion, it allows visitors to discover, in a pleasant and enriching way, places that are often inaccessible. The application not only helps to understand the history of the quarries but also enables users to grasp their significance in the development of past societies and their environmental impact. By relying on innovative digital tools, this initiative contributes to modern cultural mediation, accessible to all and particularly suited to new generations, showing curiosity for it, and persons with reduced mobility (PRM). It demonstrates that immersive technologies can play a key role in preserving and spreading collective memory, while meeting the expectations of an increasingly curious public sensitive to heritage issues.

The exhibition, accompanied by the virtual reality application, will be moved to other locations and events, allowing for reaching a broad audience. After its launch during the Heritage Days, the exhibition can be set up in other cities or historical sites, thus providing increased visibility to these underground quarries and their heritage. Moreover, the software base of the application has been designed to be modular, allowing for easy adaptation to other heritage sites or themes, ensuring its flexible reuse for other cultural mediation projects. This approach supports the application sustainability and expansion, while diversifying the experiences offered to visitors and enriching the cultural heritage offering at both the national and international levels.

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